

# Three Mechanisms Inflate Reported Performance on a Widely Used Public Chronic Kidney Disease Benchmark

Author Name

Department

Institution

City, Country

email@institution.edu

**Abstract**—Machine-learning studies on the public UCI chronic kidney disease (CKD) benchmarks routinely report accuracy at or near 100%. We ask what produces those numbers. Under repeated nested cross-validation with programmatically enforced leakage control, we identify and quantify three distinct mechanisms, none of which is predictive ability. First, *target leakage*: the released file contains an exact copy of the outcome, a post-diagnosis stage label, and the diagnostic eGFR value; including them takes every feature configuration to ROC-AUC 1.000. Second, and larger, *case-mix composition*: with every prohibited column removed, models still separate all 200 patients out of fold, because 84% of cases carry stage 3–5 disease and haemoglobin alone reaches univariate ROC-AUC 0.968; in an exploratory, post hoc and statistically underpowered subgroup of the 21 early-stage cases, sensitivity falls from 0.969 to 0.905, a hypothesis-generating signal requiring confirmation in a larger prospectively defined cohort. Third, and least visible, *pseudo-external validation*: a record-level provenance check shows that the 2015 and 2020 UCI releases, treated in the literature as independent cohorts, share their patients. All 200 patients of the 2020 release match into the 2015 release (maximum bipartite match fraction 1.000 against a permutation null of  $0.003 \pm 0.003$ ), 187 of them uniquely, with zero contradictions across ten categorical variables that took no part in the matching. Of 13 surveyed studies, three validate across these releases as though independent and one merges them into a single training set. Recovering the original measurements further shows that the published discretisation destroys 0.266 ROC-AUC of univariate signal in serum creatinine while changing no multivariable conclusion, and that 339 missing cells were silently filled with constants whose absence is strongly outcome-associated. Finally, as a *transportability probe* rather than a definitive external validation, we transfer the models unchanged to an independent critical-care cohort—the only registered source carrying the same measurements on different patients—where discrimination falls from 0.997 to 0.70 (95% CI 0.59–0.80) and Brier rises from 0.014 to 0.42. With 20 cases in a markedly different clinical setting, this difference cannot be attributed to internal optimism alone: small-sample uncertainty, case mix, feature definition, missingness and genuine distribution shift all plausibly contribute. We release the full pipeline, including a reusable provenance gate that blocks non-independent cohorts from supporting external claims, and a diagnostic protocol that would have flagged this benchmark before any model was fitted.

**Index Terms**—chronic kidney disease, data leakage, spectrum bias, transportability, external validation, benchmark quality, clinical prediction models, reproducibility

## I. INTRODUCTION

Chronic kidney disease affects roughly 9% of the world’s population and caused an estimated 1.2 million deaths in 2017, with the burden falling disproportionately on regions where diagnostic laboratory capacity is limited [1]. CKD is defined by abnormalities of kidney structure or function persisting beyond three months, operationalised principally through estimated glomerular filtration rate (eGFR) and albuminuria [2]. Because early CKD is largely asymptomatic, detection depends on testing rather than presentation.

That clinical importance is why the public UCI CKD datasets have become standard benchmarks: they are small, tidy, and answer a question that matters. It is also why results reported on them deserve scrutiny. A *benchmark on which almost every method reports near-perfect accuracy has stopped discriminating between methods*, and the interesting question becomes what, in the data, produces the ceiling.

A structured survey of prior work on this dataset and its parent release (Section II) finds every independently verified headline metric at or above 99% accuracy, 9 of 13 studies coded as using serum creatinine or eGFR as model inputs, and exactly one with a validation cohort that is genuinely external. Such numbers are more often a property of the data than evidence of clinical usefulness.

This paper began as an ordinary applied study of whether inexpensive variables—history, examination, a urine dipstick—could predict CKD status. It was built with leakage control enforced in code and repeated nested cross-validation. That construction produced an uncomfortable result: every valid configuration sat at the discrimination ceiling, with no prohibited column involved. An automated tripwire (“perfect performance means a bug”) fired, and the investigation into why became the study.

### A. Contributions

- 1) We identify and separately quantify **three mechanisms** that drive measured performance on this benchmark toward 1.0, and show that the largest is not the one the literature discusses.

- 2) We provide **record-level evidence** that two UCI releases treated as independent cohorts share their patients, invalidating published cross-dataset validations between them, and release the provenance check as reusable, dataset-agnostic machinery.
- 3) We **recover the original continuous measurements** for the analysed cohort, making the cost of the published discretisation measurable *within-cohort*, and in doing so surface undocumented constant imputation in the released file.
- 4) We report an **honest uncertainty interval** obtained by bootstrapping the entire nested procedure, which is  $1.7\text{--}2.7\times$  wider than the conventional prediction-level interval.
- 5) We report a **transportability probe** on the one independent source carrying the same measurements, in which discrimination near 1.0 falls to 0.70 on patients the models never saw. With 20 cases in a different clinical setting it is a stress test rather than a definitive external validation, and we are explicit about the several mechanisms that could produce the difference.
- 6) We propose a **diagnostic protocol** for deciding whether a benchmark can distinguish methods at all, and report a negative result about one of its own components.

## II. RELATED WORK AND PRIOR RESULTS ON THIS BENCHMARK

Recent surveys of machine learning for CKD [10] note that the majority of studies rely on the UCI-2015 release [4], and that many include serum creatinine or eGFR as inputs—a practice Kabir *et al.* describe as introducing “information leakage and circular reasoning,” since those biomarkers are themselves the clinical criteria defining the outcome.

We assembled a structured survey of 13 studies using this dataset or its parent release, recording for each the dataset version, the reported headline metric, whether post-diagnosis columns were used, and the validation design. Every cell states how it was verified. Three studies were verified from full text or abstract; the remaining ten carry an explicit `todo_hand_check` flag because paywalls blocked verification, and their coding is *attributed* to a published comparison table [10] rather than confirmed. Table I summarises.

Among the verified studies, Islam *et al.* [11] report 99.16% accuracy on the 2015 release with serum creatinine among the predictors and no external validation, and Hossain *et al.* [12] report 99.5% accuracy after *merging* the two releases into a single training set, with reported accuracies reaching 100% in cross-release validation. Both figures are consistent with the mechanisms identified below; the second is directly affected by the cohort-overlap finding of Section V-C.

The survey is a targeted sample seeded from one paper’s reference list. It is sufficient to establish that near-perfect results and cross-release validation occur; it is not sufficient to estimate how often, and we do not claim otherwise.

TABLE I  
STRUCTURED SURVEY OF PRIOR WORK ON THIS BENCHMARK ( $N = 13$ )

| Quantity                                  | Value    |
|-------------------------------------------|----------|
| Studies surveyed                          | 13       |
| Independently verified by us              | 3        |
| Awaiting hand verification (paywalled)    | 10       |
| Verified headline metrics $\geq 0.99$     | 2 of 3   |
| Coded as using serum creatinine or eGFR   | 9        |
| Using the 2020 release or a merge of both | 4        |
| Cross-validating between the two releases | 3        |
| <b>With a genuinely external cohort</b>   | <b>1</b> |

## III. DATA

### A. The analysed release

We analyse `ckd-dataset-v2.csv`, 200 patient records in 29 columns, distributed by the UCI Machine Learning Repository as *Risk Factor Prediction of Chronic Kidney Disease* [5]. The outcome `class` is 128 CKD (64.0%) and 72 non-CKD. Two rows immediately following the header are metadata, not observations, and are removed in a single place in the codebase.

With 72 minority events and 25 candidate predictors in the full valid configuration, the dataset provides 2.88 events per predictor—far below accepted minimums for prediction-model development [6]. The study is therefore underpowered for model development by construction and is presented as a measurement exercise on the benchmark itself.

### B. Two properties that shape everything

**The data are pre-discretised.** Every continuous variable was binned into interval strings before release (`sg` as  $1.019 - 1.021$ , `bgr` as  $< 112$ ). We encode each label by a single representative value—the midpoint of a closed bin, the finite edge of an open one. This mapping is a pure function of one cell’s text: it uses no cross-row statistics and no outcome information, so it cannot transfer information between folds, which is why it may legitimately be applied before splitting.

**The documented provenance is contradicted by the data.** The release documents collection at a named hospital in Bangladesh. Section V-C shows that all 200 of these patients appear in the 2015 UCI release [4], documented as an Indian hospital. We report the measurement and take no view on how the discrepancy arose; readers should treat the stated site, country and year as unverified. No claim in this paper rests on them.

### C. Prohibited columns

Three columns cannot be legitimate predictors. `affected` is an exact one-to-one copy of the outcome (Cramér’s  $V = 1.0$  across all 200 rows). `stage` is a post-diagnosis staging label, 100% CKD at stages `s3` and `s5`. `grf` is the eGFR value from which staging is banded and by which CKD is defined. A model given these is performing lookup, not prediction: at the moment of assessment none of the three exists yet.

## IV. METHODS

### A. Feature configurations

Six configurations are compared. `leaky_model` (28 variables) is *deliberately invalid*: it adds the three prohibited columns and exists solely to quantify their effect. It is labelled INVALID in every table and figure and is never quoted outside the leakage audit. The valid configurations are `full_valid` (25), `laboratory` (14), `low_cost` (11: history, examination, urine dipstick), `clinical_only` (8), and `low_cost_plus_urine_micro` (15). Machine-checked invariants hold: `clinical_only`  $\subset$  `low_cost`  $\subset$  `full_valid`, and `low_cost`  $\cup$  `laboratory` = `full_valid`.

The dataset ships no data dictionary. Ten variables whose meaning could not be settled from the file are flagged *uncertain* rather than given invented interpretations. The most consequential is `ane` (anaemia), which could be recorded clinically at no cost or read from a blood count; because its provenance is unresolved it is excluded from the low-cost configuration. This is the conservative choice: it can only *understate* low-cost performance.

### B. Leakage prevention, enforced in code

Prevention operates in three independent layers. *Declaration*: a single module names every prohibited column and each configuration’s exact contents. *Runtime guard*: every pipeline begins with a guard that raises at `fit` **and** at `transform`, so a prohibited column smuggled in at prediction time also fails; it *raises rather than drops*, because silently dropping would conceal a caller’s bug. *Assertion*: a cleanliness check runs over every pipeline before any work is done. The key adversarial test hands a valid pipeline the *entire* dataframe, including all prohibited columns, and requires it to refuse rather than quietly select its own subset.

### C. Validation design

Models are evaluated under 5-fold outer, 4-fold inner nested stratified cross-validation repeated 5 times, giving 25 independent outer test sets and 108,000 out-of-fold predictions. The nested design is used because selecting hyper-parameters and estimating performance on the same folds is a known source of optimism [7]. Six model families are compared: a prevalence dummy, penalised logistic regression, random forest, RBF support vector machine, XGBoost, and an explainable boosting machine (EBM) [9] configured as a pure additive model. Imputation, scaling, hyper-parameter selection, calibration and threshold selection all occur strictly within training folds. Uncalibrated, Platt and isotonic probabilities are compared. Confidence intervals come from a stratified patient-level bootstrap (2000 resamples); Section V-G replaces these with an interval for the whole procedure.

Calibration slope and intercept follow the standard logistic recalibration framework [8]. Two design details are worth stating because they change reported numbers. Calibration statistics are computed *within* each repeat and then summarised: averaging a patient’s probability across repeats shrinks it

TABLE II  
LEAKAGE AUDIT. BEST MODEL PER CONFIGURATION, UNCALIBRATED, POOLED OUT-OF-FOLD PREDICTIONS.

| Configuration    | $k$ | ROC-AUC (95% CI)    | Sens. | Spec. |
|------------------|-----|---------------------|-------|-------|
| Leaky (INVALID)  | 28  | 1.000 (1.000–1.000) | 1.000 | 1.000 |
| Full valid       | 25  | 1.000 (1.000–1.000) | 0.977 | 1.000 |
| Laboratory       | 14  | 0.995 (0.988–0.999) | 0.969 | 0.972 |
| Low-cost + micro | 15  | 0.992 (0.980–0.999) | 0.961 | 1.000 |
| Low-cost         | 11  | 0.992 (0.981–0.999) | 0.961 | 1.000 |
| Clinical only    | 8   | 0.953 (0.919–0.979) | 0.914 | 0.986 |

toward the centre and biases the slope upward. And complete separation is *detected* rather than optimised through—under separation the logistic recalibration slope has no maximum-likelihood solution, and an optimiser still returns a number (values in the hundreds were observed before the check was added), so such slopes are reported as undefined.

### D. The provenance gate

Before any dataset may support an external claim, it passes a record-level check. Each patient in the analysed file is a vector of intervals over 13 shared variables. A candidate record is *compatible* with a patient if the outcome matches and every shared value falls inside the corresponding interval; missing values are treated as compatible, which can only *raise* the match fraction and is therefore the conservative direction for a gate. We compute the maximum bipartite matching between patients and candidate records under this relation, and calibrate the observed matching against permutations that shuffle each candidate column independently, preserving every marginal distribution while destroying cross-variable structure.

The verdict rule—INDEPENDENT, OVERLAPPING, or SAME-SOURCE—is mechanical and was fixed in code before any external result existed. An automated test enforces that no dataset lacking an INDEPENDENT verdict can appear in any external-validation table.

## V. RESULTS

### A. Mechanism 1: target leakage

Table II gives the best model per configuration on pooled out-of-fold predictions.

The invalid configuration reaches ROC-AUC 1.000 with sensitivity and specificity both 1.000. Measured against the full valid configuration the absolute gain is 0.000—but this is a ceiling artefact, not evidence that leakage is harmless, because that baseline is itself already at 1.000. Measured against baselines that retain headroom, leakage closes **100%** of the remaining error:  $0.992 \rightarrow 1.000$  for the low-cost set and  $0.953 \rightarrow 1.000$  for the clinical-only set. Leakage takes *any* configuration to the ceiling.

That the honest baseline is already saturated is itself the finding that motivates the next section.

TABLE III

UNIVARIATE SEPARABILITY OF INDIVIDUAL PREDICTORS. NO MULTIVARIABLE LEARNING IS REQUIRED TO SEPARATE GROUPS THIS FAR APART.

| Predictor | Univar. ROC-AUC | In shared range |
|-----------|-----------------|-----------------|
| hemo      | 0.968           | 47%             |
| pcv       | 0.949           | 51%             |
| sg        | 0.888           | 58%             |
| rbcc      | 0.876           | 84%             |
| al        | 0.828           | 58%             |

### B. Mechanism 2: case-mix composition

With every prohibited column removed, leakage prevented by a guard that raises, and an automated audit confirming no violation across all 36 pipelines actually run, the full valid configuration *still* separates all 200 patients out of fold. This is a property of the sample.

Two facts explain the ceiling. First, single predictors already separate the groups: haemoglobin alone achieves univariate ROC-AUC 0.968 (Table III). Second, the cohort is dominated by advanced disease: 107 of 128 CKD patients (84%) are staged s3–s5, leaving only 21 early-stage cases. This is the profile of a case-control-like contrast between established kidney failure and comparatively healthy controls—not of a consecutive screening series, in which most true cases would be early, asymptomatic and biochemically near-normal. Anaemia, low haematocrit and abnormal urine concentration are *consequences* of established kidney disease; they are precisely the features absent in the patients a screening programme most needs to identify.

Restricting to early CKD (s1–s2) versus non-CKD, sensitivity at the prespecified threshold falls from 0.969 to 0.905 for the low-cost configuration and from 0.891 to 0.762 for the clinical-only configuration (Fig. 1).

**These early-stage figures are exploratory and must not be read as estimates.** The subgroup contains **21 cases**, the analysis was defined post hoc after the ceiling was observed rather than prespecified, and no formal comparison is made or implied: with 21 cases the confidence interval on any subgroup sensitivity spans a large part of the achievable range, and a difference of the size reported would not survive a demand for statistical significance. We report the numbers as *hypothesis-generating* for three reasons: the direction is consistent across every cheaper configuration, the underlying case-mix imbalance (107 of 128 cases at s3–s5) is an arithmetic fact about the sample rather than an inference, and omitting the analysis would leave the aggregate figures looking more like screening performance than they are. Confirmation requires a larger, prospectively defined early-stage cohort. We draw no causal conclusion and make no claim that the effect size generalises beyond this sample.

### C. Mechanism 3: pseudo-external validation

The two mechanisms above concern a single dataset. The third concerns how the dataset is used. Several published

TABLE IV

UNIVARIATE SEPARABILITY, BINNED VERSUS RECOVERED CONTINUOUS VALUES, EVALUATED ON THE SAME PATIENTS. SELECTED VARIABLES.

| Variable | Binned | Continuous | Lost to binning |
|----------|--------|------------|-----------------|
| sc       | 0.680  | 0.946      | +0.266          |
| pot      | 0.518  | 0.588      | +0.070          |
| sod      | 0.806  | 0.820      | +0.014          |
| hemo     | 0.985  | 0.985      | −0.000          |
| sg       | 0.913  | 0.913      | 0.000           |

studies train on the 2015 UCI release and validate on this file—distributed separately, with a different stated collection site, sample size and year—and present the result as external validation. That inference requires the two releases to contain different patients.

They do not. Applying the gate of Section IV-D (Fig. 2):

- **All 200** patients match into the 2015 release; the maximum bipartite match fraction is **1.000**, against a permutation null of **0.003 ± 0.003** (maximum 0.015 over 200 shuffles).
- **187 of 200** patients are compatible with exactly *one* record; the maximum anywhere is two.
- Across those 187 uniquely pinned pairs, **ten categorical variables that took no part in the matching agree with zero contradictions** ( $\approx 1,500$  comparisons): rbc/pc 1 ↔ abnormal, pcc/ba 1 ↔ present, htn/dm/cad/pe/ane 1 ↔ yes, appet 1 ↔ poor.

A coincidental alignment does not reproduce ten unused variables. The analysed file is a discretised subset re-release of the 2015 dataset. Of the 13 surveyed studies, three validate across the two releases as though they were independent cohorts, and one merges them into a single training set before reporting accuracy. On this evidence those procedures evaluate models on their own training population.

This is not a criticism of those authors. Nothing in either dataset’s documentation indicates the overlap, and the present study registered the 2015 release as its *own* primary external-validation candidate before the gate refused it.

### D. What the discretisation destroyed

Because the overlap is with a continuous-valued release of the *same patients*, the cost of the published binning is measurable within-cohort: no population shift, no case-mix difference, no sampling variation. Recovery is restricted to the 187 uniquely pinned patients, and every recovered value is verified to lie inside its own patient’s published interval.

The loss is concentrated in one variable, and it is the diagnostically decisive one (Table IV, Fig. 3). Serum creatinine loses **0.266** ROC-AUC; every other variable loses at most 0.070. The published bins explain why: the interval  $< 3.65$  absorbs 137 of 177 patients and spans 0.5–3.6 mg/dL—from unambiguously normal (0.6–1.2) through severe renal impairment—and is only 52% CKD, while every other bin is 100% CKD. In this release serum creatinine is not a graded

Figure R12. Case-mix analysis: why near-ceiling discrimination here is not a screening result (EXPLORATORY)

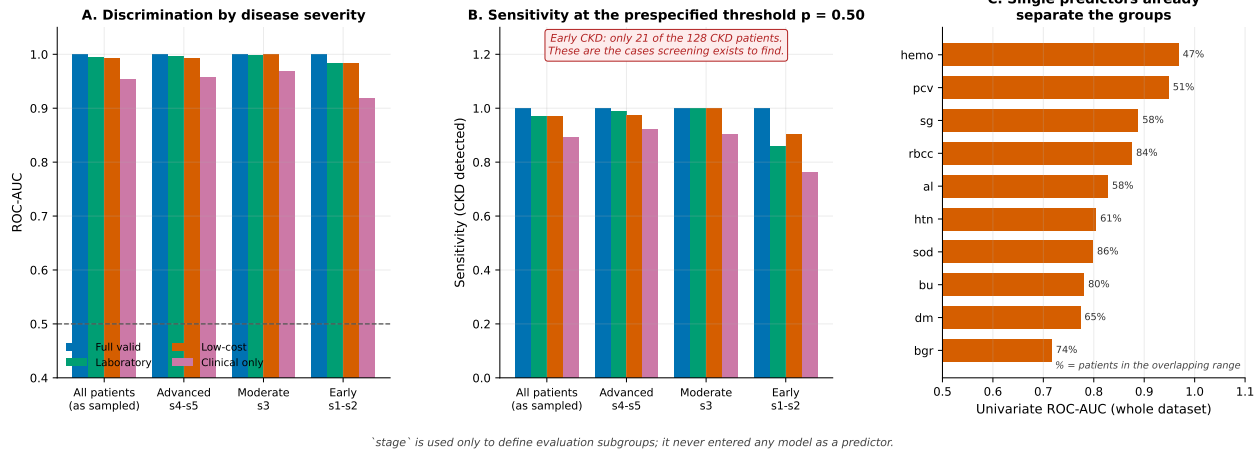


Fig. 1. Performance by disease-stage subgroup. Discrimination measured on the sample as a whole appears optimistic relative to the early-stage patients that screening exists to find; the early subgroup contains only 21 cases, so these comparisons are exploratory and hypothesis-generating rather than estimates. Panel C shows that single predictors already separate the groups, which is why no multivariable learning is required to reach the ceiling.

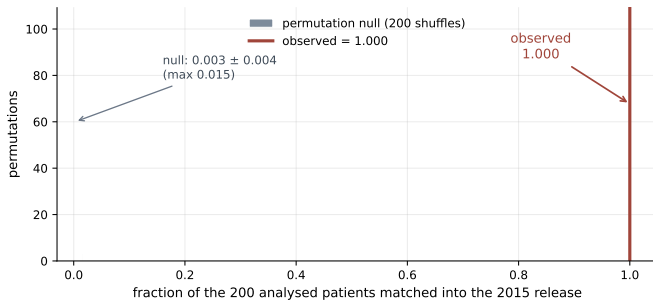


Fig. 2. Cohort-overlap evidence. The observed match fraction (1.000) against a permutation null that preserves every marginal distribution while destroying cross-variable structure. A coincidental alignment does not reproduce ten variables that played no part in the matching.

measurement but a coarse flag that fires only once creatinine is already extreme.

This has a consequence for where the leakage boundary should be drawn. We excluded *grf* as a post-diagnosis derivative but retained serum creatinine, on the grounds that it is a routinely measured analyte rather than a diagnostic label. On the released data that judgement is comfortable *because binning has flattened the variable*. On the underlying measurements it is much less so: creatinine alone reaches 0.946, close to the quantity that defines the outcome. A study using the continuous release would need to defend that inclusion far more carefully—and could not learn this from the released file alone.

#### E. Undocumented constant imputation

The recovery surfaced a second property of the release. The analysed file contains exactly one missing cell; its source contains many. Across the recoverable variables, **339** cells that the source leaves blank carry values here, and for **all 13**

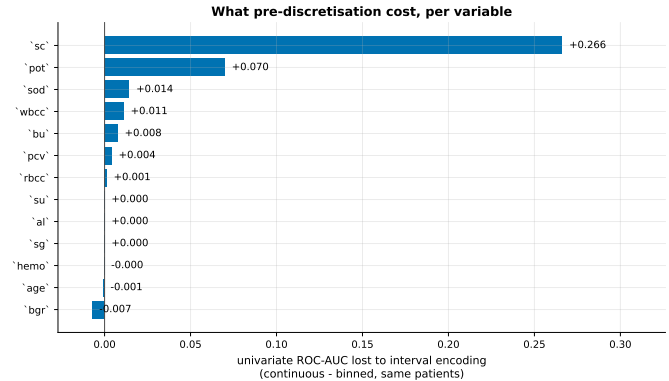


Fig. 3. Univariate ROC-AUC lost to the published interval encoding, per variable, on the same patients. The loss is negligible everywhere except serum creatinine.

variables every such cell was filled with a *single constant*—in each case the clinically normal range (*sod* → 133 – 138, *rbcc* → 4.46 – 5.05, *al* → zero).

The missingness is not random. A patient with CKD is 4.9 to over 30 times more likely to have these measurements absent from the source, which is what one expects when tests are ordered selectively. Filling every such cell with the normal value therefore assigns normal-looking laboratory results to precisely the patients most likely to be ill, and does so invisibly: nothing in the released file distinguishes a measured normal result from a supplied one.

We state its direction plainly because it runs *against* this paper's thesis. Every other mechanism examined here inflates measured performance; this one deflates it, since substituting normal values for the sickest patients makes the classes harder to separate. The implication is not that the benchmark is harder than it looks—it is that a substantial minority of some columns are not measurements at all, which no analysis of the released

TABLE V  
PREDICTION-LEVEL VERSUS WHOLE-PROCEDURE 95% INTERVALS FOR  
ROC-AUC. 200 RESAMPLES PER CELL, NO FAILURES.

| Config.    | Model | Pred.-level | Proc.-level | Ratio |
|------------|-------|-------------|-------------|-------|
| Laboratory | SVM   | 0.988–0.999 | 0.970–0.999 | 2.7×  |
| Low-cost   | RF    | 0.981–0.999 | 0.968–0.999 | 1.7×  |
| Low-cost   | SVM   | 0.979–0.999 | 0.960–0.998 | 1.8×  |
| Full valid | SVM   | degenerate  | 0.999–1.000 | —     |

file could discover.

#### F. Would the continuous data have changed any conclusion?

We modelled the same 187 patients twice under identical design, seeds and pipelines, differing only in the representation of the recoverable variables. The largest difference in either direction is **0.0122** ROC-AUC—an order of magnitude inside the intervals of Section V-G. Nothing changes.

That null is more informative than a difference would have been. Serum creatinine gains 0.266 ROC-AUC on its own when its true values are restored, and the multivariable models do not benefit at all, because the ceiling means that information is already present several times over via haemoglobin and packed cell volume. *One can destroy the resolution of the most diagnostic laboratory analyte in the dataset and the models will not notice.* What this rules out is the specific worry that the published binning explains the benchmark’s ceiling. It does not; the case mix does.

#### G. Honest uncertainty

Conventional intervals resample patients over pooled out-of-fold predictions while holding the *fitted models* fixed, and therefore cannot see the variability introduced by model selection, tuning and calibration being re-made on a different sample. We removed that shortcut: for each resample the entire nested procedure is re-run from scratch, 200 times per cell.

Within each resample the outer folds are formed over *patients* rather than rows, so every copy of a duplicated patient stays on the same side of every split. Without that constraint a model would be evaluated on patients it had trained on—the failure this paper is about, and not excusable in its own methods.

The honest intervals are 1.7–2.7× wider (Table V). These are the intervals a reader should use when asking whether two configurations differ; on this evidence, the study’s inability to distinguish the low-cost and laboratory sets is if anything *understated* by the narrower intervals conventionally reported.

#### H. Robustness

Table VI summarises analyses that test whether the conclusions depend on specific choices. None does.

The label-robustness asymmetry deserves note because it stresses the paper’s own supporting claim: the four disputed patients are precisely those whose history, examination and dipstick findings look unremarkable while their laboratory values indicate advanced disease. If their `notckd` labels are

TABLE VI  
ROBUSTNESS ANALYSES

| Question                                                | Answer                                                                                                                  |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Do the 4 internally contradictory labels drive results? | Deleting: no ( $ \Delta  \leq 0.0025$ ). <b>Flipping</b> : costs low-cost up to <b>0.0241</b> vs 0.0022 for laboratory. |
| Is the midpoint encoding load-bearing?                  | No. Four encodings incl. one-hot (86 columns); largest deviation <b>0.0101</b> .                                        |
| How much is internal validation already correcting for? | Largest apparent-minus-nested optimism <b>+0.0072</b> .                                                                 |
| Does isotonic really beat Platt at $n = 200$ ?          | Yes. <b>105/120</b> paired wins (88%) at 100 outer test sets.                                                           |

the errors, they are exactly the patients a low-cost instrument would miss—and a laboratory panel would not.

The calibration result is worth one further line, since it contradicts the usual small-sample expectation. Across 100 outer test sets, isotonic calibration achieved median Brier 0.0288 against 0.0337 for Platt scaling and 0.0345 uncalibrated, with median calibration slope 1.148 against 2.378 and 1.887 respectively.

#### I. Supporting result: cost-stratified feature sets

Under leakage-controlled validation, 11 variables obtainable from history, examination and a urine dipstick reach ROC-AUC 0.992 (95% CI 0.981–0.999) against 0.995 for the 14-variable laboratory panel—a difference well inside either interval, and unresolved at this sample size.

At identical cost, the additive EBM gives up only 0.0066 ROC-AUC to the RBF SVM (0.987 vs 0.993) while achieving a calibration slope of **1.009** against **1.257**, and yields per-variable shape functions a clinician can read and challenge (Fig. 4). On this sample the readable model is also the better-calibrated one, so the usual accuracy-versus-interpretability trade-off does not bind. One caveat is recorded rather than smoothed over: `bp` (Diastolic) runs opposite to its own marginal association—a suppression effect beside `bp limit` and `htn`—and must not be read in isolation.

**This is a statement about cost tiers on this case mix. It is not a screening result**, because a sample in which 84% of cases carry moderate-to-severe disease is not a screening population.

#### J. A transportability probe on an independent cohort

Every result above is internal. The provenance gate disqualified the obvious external candidate (Section V-C), leaving one registered source that carries the *same measurements* on different patients: the MIMIC-IV critical-care demonstration database, from a different country and health system. We extracted 100 subjects in this study’s schema—26 coded CKD

Low-cost EBM shape functions (mean  $\pm$  SD across outer folds)

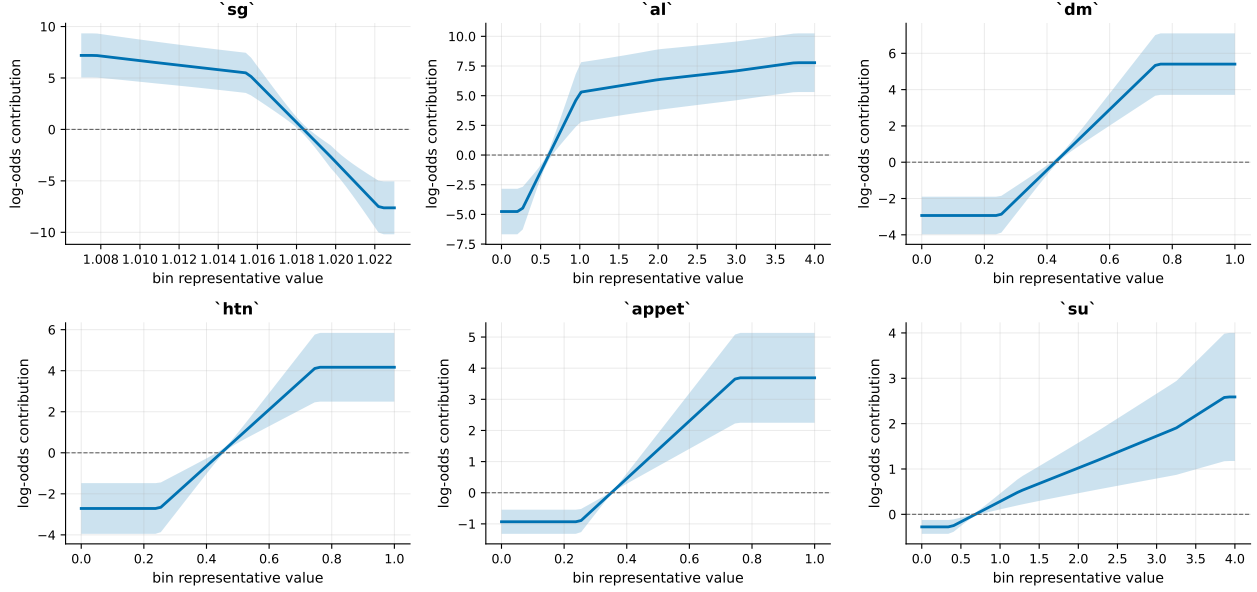


Fig. 4. Fitted shape functions of the low-cost additive model, averaged over 25 outer folds (shaded band:  $\pm 1$  SD). Directions are clinically coherent: low urine specific gravity raises risk, albuminuria raises it steeply, and diabetes, hypertension and poor appetite all push the same way.

by ICD-10 N18\* or ICD-9 585\*, with nine blood analytes at 92–94% coverage plus urine specific gravity.

We describe this experiment as a *transportability probe* or stress test, not as an external validation of the kind that would support a performance claim. A validation cohort would need to be adequately powered, drawn from a population in which the model is intended to operate, and labelled by the same standard as the development data. This cohort satisfies none of those conditions: it contributes 20 cases, comes from critical care, and is labelled administratively. What it can do—and what a validation cohort is not needed for—is test whether performance near the ceiling persists when the patients change at all.

Three properties of the extraction are load-bearing. The derived label is named so that the registry’s prohibited-column declaration, written before the extraction existed, already covers it. A *temporal guard* discards laboratory values that do not strictly precede the admission at which CKD was first coded, removing 8,754 values. And the six subjects left with no eligible measurement are reported rather than imputed into existence.

Models were trained on all 200 analysed patients and applied **without any refitting**. The transfer feature set is the intersection of the two schemas—ten variables—passed as a “feature override” on the existing valid configuration so that both leakage guards still run. An internal reference is computed on exactly that feature set, so the drop caused by transfer is separable from any drop caused by using fewer variables.

**Discrimination falls sharply on the new cohort.** Internal ROC-AUC on this feature set is 0.997; on the probe cohort it

TABLE VII  
FROZEN TRANSFER TO AN INDEPENDENT COHORT. INTERNAL REFERENCE AND EXTERNAL RESULT USE THE SAME TEN VARIABLES.

| Model         | Int. AUC | Ext. AUC (95% CI)   | Brier | Sens. | Spec. |
|---------------|----------|---------------------|-------|-------|-------|
| Logistic      | 0.997    | 0.702 (0.594–0.794) | 0.409 | 0.95  | 0.47  |
| SVM           | 0.997    | 0.699 (0.599–0.788) | 0.423 | 0.95  | 0.45  |
| Random forest | 0.997    | 0.708 (0.605–0.803) | 0.418 | 0.95  | 0.39  |

is 0.699–0.708, a fall of roughly 0.29 (Table VII). All three model families agree, and the bootstrap intervals exclude the internal estimate by a wide margin. The inference this supports is a conditional one: performance measured near the ceiling on this benchmark is not, on this evidence, a reliable guide to performance on a different patient population. It does not by itself establish how much of the internal figure was optimism, and we quantify neither.

**Calibration fails worse, and for a legible reason.** External Brier scores are 0.409–0.423 against 0.014 internally, with calibration intercepts far below zero: the models assign high risk to almost everyone. At the prespecified threshold they retain sensitivity 0.95 while specificity collapses to 0.39–0.47. Prevalence is the obvious culprit—64% in training against 21% externally. Applying recalibration-in-the-large, the minimal correction a deployer would make, reduces the Brier score to 0.172–0.175. It cannot change the ranking, so discrimination is unmoved by construction, and it leaves no prediction above 0.50, so the model flags nobody at the prespecified threshold. *The binding constraint is the discrimination of 0.70, not the calibration*, and no post-hoc correction addresses it.

**The difference cannot be attributed to internal optimism**

**alone.** This design cannot decompose it, and at least six mechanisms plausibly contribute:

- *Small-sample uncertainty.* The probe rests on 20 cases and 74 controls; the interval spans 0.59–0.80, so the point estimate is itself imprecise.
- *Case-mix differences.* The development sample is 84% stage 3–5 with 64% prevalence; the probe cohort is critical-care with 21% prevalence and a different severity profile.
- *Measurement and feature definition.* Analytes are drawn from hospital laboratory systems under different assay and reporting conventions, and the urine dipstick tier is semi-quantitative text mapped to an ordinal scale rather than natively recorded.
- *Missingness differences.* The development file contains one missing cell because the release imputed the rest with constants (Section V-E); the probe cohort has genuine missingness imputed within training folds. The two arms therefore differ in how absent measurements are represented.
- *Clinical setting.* Critical-care admission selects patients on acute physiology, which perturbs precisely the analytes the models rely on.
- *Genuine population and distribution shift,* including differences in country, health system, era and coding practice.

Internal optimism is a seventh candidate, and the study offers indirect support for it—the honest intervals of Section V-G and the saturation behaviour of Section VI both indicate a sample that should not be expected to transfer—but the probe does not isolate it. Attributing the fall to any single mechanism, optimism included, would overstate what a 20-case comparison can support.

None of this diminishes what the probe contributes. It is the only evidence in the study drawn from patients the models had never seen, it is the only such evidence the registered sources permit once the provenance gate has excluded the alternative, and its direction is unambiguous and consistent across three model families. A benchmark whose reported discrimination is 0.99 and whose models score 0.70 on the first independent cohort available warrants the caution this paper argues for, whatever the decomposition of that gap turns out to be.

## VI. A PROPOSED DIAGNOSTIC FOR BENCHMARK INFORMATIVENESS

Every mechanism above was invisible in the number the literature reports. That is not an accident of this dataset. Discrimination answers “how well did this model separate these patients?” and says nothing about whether the dataset can distinguish good methods from bad ones. This section proposes three measurements that address the gap and applies them here.

We state the provenance of the idea plainly: none of the three components is new. Direct standardisation is routine in epidemiology, learning curves are long established, and comparing a model against a strong single-feature baseline

TABLE VIII  
MULTIVARIABLE LIFT OVER THE BEST SINGLE PREDICTOR.

| Configuration   | Model | Best single | Lift (95% CI)         |
|-----------------|-------|-------------|-----------------------|
| Low-cost (11)   | 0.993 | sg 0.888    | +0.105 (0.071, 0.145) |
| Laboratory (14) | 0.994 | hemo 0.968  | +0.026 (0.010, 0.043) |
| Full valid (25) | 1.000 | hemo 0.968  | +0.032 (0.014, 0.053) |

is ordinary practice. What is proposed is their composition into a *benchmark-level* check that can be run before a dataset is trusted, with reporting thresholds. Those thresholds are calibrated on one benchmark and are offered for discussion, not as standards.

### A. Multivariable lift

How much did modelling add over the single most separable column? We report  $AUC(\text{model}) - AUC(\text{best single predictor})$ , with a paired patient-level bootstrap interval. The baseline is given the benefit of the doubt: a single raw column has no fitted direction, so its AUC is taken as  $\max(AUC, 1 - AUC)$ , which makes the lift smaller rather than larger.

The laboratory configuration—fourteen variables, a full blood panel and urine microscopy—improves on haemoglobin *alone* by 0.026 ROC-AUC (Table VIII). The full valid configuration, with 25 variables, adds 0.032. A benchmark on which the entire modelling exercise is worth three hundredths of an AUC over one raw measurement cannot rank methods, whatever numbers it produces. The low-cost set is the informative exception at +0.105: it is the one place in this study where combining variables demonstrably works, which is consistent with its variables being non-redundant rather than proxies for one another.

### B. Saturation

How much of the training data was needed? Using 10% of each training fold—roughly 16 patients—already reaches ROC-AUC 0.970, against 0.996 on the full fold. Ninety per cent of the achievable gain above chance arrives at 10% of the data (Fig. 5). A benchmark solved by a tenth of its training examples cannot reward sample efficiency, and its reported numbers describe the task rather than the learner.

### C. Case-mix standardisation, and a negative result

What would the numbers be in a population with a stated severity mix? We re-weight cases to the case mix of a published community screening series [10] (22% stage 1, 46% stage 2, 32% stage 3) and recompute. Standardising the AUC moves it by at most 0.008—so on this evidence it would not flag a benchmark that Section V-B shows is case-mix dependent at a threshold.

The explanation is specific to this dataset, and stating it precisely matters. AUC is a rank statistic, so re-weighting which cases are present moves it only insofar as it changes whether cases outrank controls. Here even the early-stage cases are well separated—their subgroup ROC-AUC is 0.983, on 21

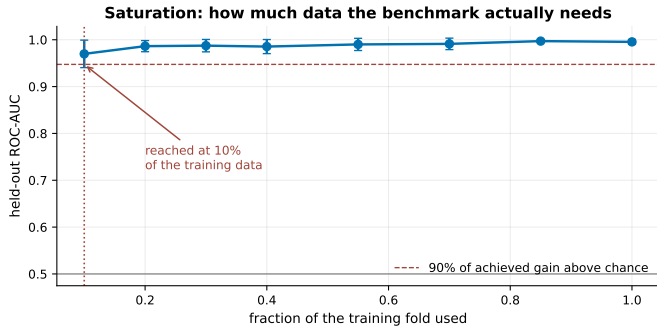


Fig. 5. Held-out discrimination against the fraction of the training fold used. The curve is flat from a tenth of the data onward, so the benchmark cannot distinguish methods by how efficiently they learn.

cases and therefore imprecise—so shifting weight toward them changes few rankings. Sensitivity at a fixed threshold has no such invariance: a case below the threshold is missed however it ranks, and the same re-weighting shifts it by up to 0.056, an order of magnitude more, flagging the laboratory configuration. Because the re-weighting concentrates on a 21-case stratum, the standardised quantities inherit that imprecision and are reported as exploratory; the effective case count after weighting is 38.2, and we draw no inference from the standardised values other than the qualitative contrast between the two metrics’ sensitivity to case mix.

This is **not** evidence that AUC is generally blind to spectrum effects. It plainly is not: a case mix that pushes cases below controls does move AUC, which is why spectrum bias is a recognised threat to discrimination measures. What this dataset shows is narrower and still useful—a sample can be separable enough that AUC absorbs a large change in case mix while a threshold-based metric does not. The practical consequence is that standardising discrimination alone can give false reassurance, so an operating-point metric should be standardised alongside it. Which of the two moves more will depend on the dataset.

Applied here, the protocol raises at least one flag for every configuration, and reaches that conclusion without reference to any of the three mechanisms that motivated it. Whether the thresholds transfer is an open question; the machinery is released so that others can find out.

## VII. DISCUSSION

This study set out to measure one failure mode and found three. Each pushes measured performance toward 1.0, none is predictive ability, and they compound: controlling any one leaves the others free to produce a near-perfect number.

They are also ordered by how easy each is to miss. Leakage is visible in the column list. Case mix requires examining who is in the sample. Cohort overlap requires comparing two datasets nobody had reason to suspect were one—and is invisible from inside either file, since both look like well-formed, separately documented datasets. That ordering is the

paper’s general point: a benchmark’s trustworthiness is not a property one can establish by analysing it.

Three practices follow, each cheap and each of which changed a conclusion here. *Declare prohibited columns and enforce them with something that raises rather than drops. Inspect the stage distribution and univariate separability before believing a discrimination figure*—if a single predictor reaches 0.97 alone, the modelling is not what is producing the result. *Verify that a validation cohort is a different set of patients, by record-level comparison rather than by documentation.*

The transportability probe in Section V-J lends that argument empirical support, within limits we state rather than minimise. A model whose internal discrimination is 0.997 achieves 0.70 on patients it has not seen, with calibration that would refer almost the entire cohort. The probe is underpowered at 20 cases, drawn from critical care, and labelled administratively, so the gap reflects some combination of small-sample variation, case mix, measurement and feature definition, missingness handling, clinical setting and distribution shift, alongside any internal optimism; the design cannot apportion it. The claim it supports is correspondingly modest and is the one the paper needs: on this benchmark, an internal figure near 1.0 is a poor guide to performance on different patients. Establishing *why*, and estimating what performance in an intended setting would be, requires an adequately powered cohort from that setting.

The wider implication concerns how benchmark results are read. Two of the three mechanisms here are properties of the data that no amount of careful modelling can correct, and the third invalidates the evidence normally taken as decisive. A reported accuracy of 99% on this benchmark is compatible with a model that has learned nothing transferable, and distinguishing the cases requires exactly the leakage audit, subgroup analysis and provenance check reported here. Section VI asks the further question of whether the benchmark can rank methods at all, and answers no: the entire modelling exercise is worth about 0.03 ROC-AUC over one raw measurement, and a tenth of the training data suffices.

### A. Extension to synthetic tabular and synthetic patient data

The failure modes documented here are not specific to real data, and the safeguards generalise directly to pipelines that generate or evaluate synthetic tabular records. Synthetic-data evaluation is in one respect harder: a generator trained on a source table can reproduce individual rows, so the question “is the evaluation set independent of the training set?” becomes a question about memorisation as well as provenance, and the mechanisms of Sections V-C and IV-D answer only the latter.

Seven components of the protocol we implement are already directly reusable. Cryptographic dataset fingerprints establish that an input has not silently changed; here the source file is SHA-256 checksummed on every load and re-verified in continuous integration. Exact-row and near-duplicate detection across training, validation, test and reference sets generalises the interval-containment matching of Section V-C, which is a tolerance-based near-duplicate test calibrated against a

permutation null. Entity-level overlap checks using stable identifiers, or privacy-preserving linkage where identifiers cannot be shared, correspond to the bipartite matching and held-out categorical confirmation reported there. Feature–target leakage scans and the detection of target proxies and post-outcome variables correspond to the declared prohibited-column sets and the outcome-proxy audit of Section IV-B. Enforcing that all preprocessing is fitted inside cross-validation folds is the guard tested here. Provenance manifests recording origin, version, transformations and split history correspond to the run manifest and external-dataset registry. And the fail-closed gate—blocking evaluation when overlap, leakage or undocumented provenance exceeds a threshold—is implemented as a test that fails when any non-independent cohort appears in an external-validation table.

Two further components are specific to synthetic data and are *not* implemented here. Membership-inference and memorisation checks ask whether a synthetic record discloses that a particular real patient was in the training set. Nearest-neighbour similarity and distance-to-closest-record diagnostics ask whether synthetic rows are near-copies of real ones, which is both a privacy question and an evaluation-validity question: a synthetic test set whose records sit arbitrarily close to training records reproduces, in a subtler form, exactly the pseudo-external validation of Section V-C. Adding them to the gate we release is the natural next step and we have not taken it; the present work provides the provenance half of the problem, not the memorisation half.

We note the analogy carries a caution as well as a recipe. The most consequential mechanism in this study was not leakage but case mix—something no fingerprint, hash or duplicate scan would detect. A synthetic cohort can be free of memorisation, pass every overlap check, and still be uninformative because its severity distribution makes the task trivial. Diagnostics of the kind proposed in Section VI are complementary to provenance gating for that reason, and we would expect a mature synthetic-data evaluation pipeline to run both.

## VIII. LIMITATIONS

Model development rests on 200 patients, and **the study has not adequately powered external validation**. What it has is a transportability probe (Section V-J) on 94 subjects and 20 cases. The natural comparator—the larger 2015 release—is disqualified by our own finding, and the gate enforces that rather than merely stating it. The probe cohort is drawn from critical care rather than any screening or general population, sits in a different country and health system, and carries administratively coded rather than adjudicated labels with no chronicity criterion verified. It demonstrates that discrimination measured on this benchmark does not persist on a different population; it *cannot* apportion that difference among small-sample uncertainty, case mix, measurement and feature definition, missingness handling, clinical setting, distribution shift and internal optimism, and it is far too small and too unrepresentative to estimate what performance in any intended

setting would be. No performance claim for any population should be drawn from it. The most suitable remaining candidate is a community-based series with 68% of cases at stages 1–2, access to which is restricted.

The sampling is hospital-based and single-centre; the 64% prevalence is a property of recruitment, and PPV and NPV reported at that prevalence do not transfer. The case mix is the most consequential limitation and is why no figure here estimates screening performance.

**The early-stage subgroup analysis is exploratory and underpowered.** It rests on 21 cases, was specified post hoc rather than in advance, and is reported without formal inferential comparison. Its role in the argument is to illustrate a case-mix imbalance that is established independently by arithmetic—107 of 128 cases at stages 3–5—and not to estimate early-stage performance, which 21 cases cannot support. The direction is consistent across configurations, but the magnitude should be treated as hypothesis-generating and requires confirmation in a larger, prospectively defined early-stage cohort. The same caution applies to the standardised quantities in Section VI, whose weights concentrate on that stratum.

Four records are internally contradictory, and while deleting them changes nothing, trusting their staging instead costs the low-cost configuration up to 0.0241 ROC-AUC—so this limitation is half-answered, not answered. The data are cross-sectional, so chronicity, which the definition of CKD requires, cannot be established. Age is banded and no sex, ethnicity or socioeconomic variable exists, so **subgroup and fairness analysis is impossible**—a reason for caution rather than an excuse for silence.

The literature survey is targeted rather than systematic, and 10 of its 13 rows await hand verification behind paywalls. This matters more than most limitations here, because the paper’s opening claim is a claim about the literature.

The diagnostic protocol of Section VI is a proposal, not a validated instrument. Its thresholds are set from the behaviour of a single benchmark, its components are individually familiar rather than novel, and one of them—standardised AUC—failed to flag a dataset it should have, which we report rather than remove. It should be read as a starting point for discussion.

Reproducibility has an honest limit: identical seeds reproduce identical predictions within an environment, but across an environment rebuild agreement is to approximately  $10^{-16}$  rather than bit-for-bit, because floating-point summation order depends on the installed BLAS.

We make no claim about clinical utility, causality, deployment readiness or generalisability, and none about *how* the provenance discrepancy arose.

## IX. CONCLUSION

Machine-learning results on this benchmark cluster at the discrimination ceiling. We identify three mechanisms that put them there. Target leakage takes any configuration to ROC-AUC 1.000. Case mix—84% of cases at stage 3–5, with haemoglobin alone at 0.968—puts valid models at the ceiling unaided, and is the larger effect. And the two releases the

literature treats as independent cohorts share their patients record-for-record, so published cross-dataset validations between them measure resubstitution.

Transferring the models unchanged to an independent cohort, as a transportability probe rather than a definitive external validation, then indicates what those mechanisms may cost: discrimination of 0.997 becomes 0.70, and calibration fails badly enough to refer nearly everyone. With 20 cases from a critical-care population, that difference reflects some combination of small-sample uncertainty, case mix, measurement definition, missingness, clinical setting and distribution shift as well as any internal optimism, and the probe cannot separate them. It bounds the direction rather than the magnitude—but the direction is the point.

What the work supports is a methodological claim in two parts. About this benchmark: its reported numbers are largely explained by properties of the data, the largest of which is rarely examined and the least visible of which invalidates the cross-dataset validations that would otherwise be its strongest evidence; and what remains did not transfer to the one independent cohort we could test. About method: all three mechanisms were found by checks that can be run *before* any model is fitted—declare and enforce prohibited columns, inspect the case mix and univariate separability, and verify at record level that a validation cohort is a different set of patients. We add a fourth question worth asking of any benchmark before trusting it: does modelling beat the best single column, and does the task need more than a fraction of the data? On this benchmark the answers are barely and no. Section VII-A sets out how the same safeguards, and the two memorisation-specific checks we do not implement, would apply to synthetic tabular and synthetic patient data.

Prospective validation in a genuine screening series would be required before any clinical interpretation is warranted, and no dataset examined here can substitute for it.

## REPRODUCIBILITY

The complete pipeline, including the provenance gate as reusable dataset-agnostic machinery, is available in the accompanying repository. One master seed derives every fold and model seed; the raw file is checksummed on every load and never modified; all 108,000 out-of-fold predictions are written to a single table from which every reported number is derived; and 416 automated tests enforce the leakage, determinism and report-consistency guarantees, including the provenance gate and the external-cohort temporal guard. Environment and BLAS build are recorded in a lock file. Reporting follows the spirit of TRIPOD+AI [3]; a completed checklist accompanies the repository.

## REFERENCES

- [1] GBD Chronic Kidney Disease Collaborators, “Global, regional, and national burden of chronic kidney disease, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017,” *The Lancet*, vol. 395, no. 10225, pp. 709–733, 2020. doi:10.1016/S0140-6736(20)30045-3
- [2] Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group, “KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease,” *Kidney International*, vol. 105, no. 4S, pp. S117–S314, 2024. doi:10.1016/j.kint.2023.10.018
- [3] G. S. Collins, K. G. M. Moons, P. Dhiman, *et al.*, “TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods,” *BMJ*, vol. 385, e078378, 2024. doi:10.1136/bmj-2023-078378
- [4] L. Rubini, P. Soundarapandian, and P. Eswaran, “Chronic Kidney Disease,” UCI Machine Learning Repository, 2015. doi:10.24432/C5G020
- [5] M. A. Islam and S. Akter, “Risk Factor Prediction of Chronic Kidney Disease,” UCI Machine Learning Repository, 2020. doi:10.24432/C5WP64
- [6] R. D. Riley, K. I. E. Snell, J. Ensor, *et al.*, “Minimum sample size for developing a multivariable prediction model: Part II — binary and time-to-event outcomes,” *Statistics in Medicine*, vol. 38, no. 7, pp. 1276–1296, 2019. doi:10.1002/sim.7992
- [7] S. Varma and R. Simon, “Bias in error estimation when using cross-validation for model selection,” *BMC Bioinformatics*, vol. 7, art. 91, 2006. doi:10.1186/1471-2105-7-91
- [8] B. Van Calster, D. Nieboer, Y. Vergouwe, B. De Cock, M. J. Pencina, and E. W. Steyerberg, “A calibration hierarchy for risk models was defined: from utopia to empirical data,” *Journal of Clinical Epidemiology*, vol. 74, pp. 167–176, 2016. doi:10.1016/j.jclinepi.2015.12.005
- [9] H. Nori, S. Jenkins, P. Koch, and R. Caruana, “InterpretML: a unified framework for machine learning interpretability,” arXiv:1909.09223, 2019.
- [10] M. A. Kabir, S. Munira, D. T. Azad, S. M. Ikram, M. H. R. Sarker, and S. M. A. Hanifi, “Community-based early-stage chronic kidney disease screening using explainable machine learning for low-resource settings,” *International Journal of Medical Informatics*, 2026. arXiv:2601.01119
- [11] M. A. Islam, M. Z. H. Majumder, and M. A. Hussein, “Chronic kidney disease prediction based on machine learning algorithms,” *Journal of Pathology Informatics*, vol. 14, art. 100189, 2023. doi:10.1016/j.jpi.2023.100189
- [12] M. F. Hossain, S. T. Diya, and R. Khan, “ACD-ML: Advanced CKD detection using machine learning: a tri-phase ensemble and multi-layered stacking and blending approach,” *Computer Methods and Programs in Biomedicine Update*, vol. 7, art. 100173, 2025. doi:10.1016/j.cmpbup.2024.100173